

Министерство науки и высшего образования Российской Федерации
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Факультет программной инженерии и компьютерной техники

ОТЧЁТ
О ЛАБОРАТОРНОЙ РАБОТЕ № 2
Администрирование систем и сетей
по теме:
Создание взаимосвязанной IP-сети
Желаемая оценка: 3

Выполнили: студенты группы Р3434

Р. Д. Баянов

Д. А. Кузнецов

Проверил:

преподаватель Д. Б. Афанасьев

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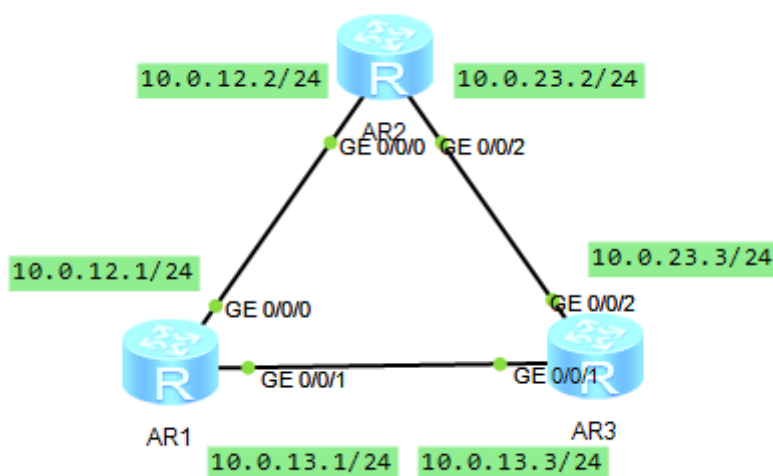
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ПЛАН РАБОТЫ

1. Настройка IP-адресов для интерфейсов на маршрутизаторах.
2. Настройка статических маршрутов для установления связи между маршрутизаторами.
3. Создание процессов OSPF на устройствах и включение OSPF на интерфейсах.
4. Настройка аутентификации OSPF.
5. Настройка OSPF для анонсирования маршрутов по умолчанию.
6. Управление выбором маршрутов OSPF на основании их стоимости.

1 Ход работы

1.1 Настройка основных параметров устройства



router1:

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname router1
[router1]
```

1.2 Вывод на экран IP-адреса текущего интерфейса и таблицы маршрутизации маршрутизатора

Вывод до конфигурации устройств:

Интерфейсы устройств:

router1:

```
[router1]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 1
The number of interface that is UP in Protocol is 1
The number of interface that is DOWN in Protocol is 3
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	up	down
GigabitEthernet0/0/1	unassigned	up	down
GigabitEthernet0/0/2	unassigned	down	down
NULL0	unassigned	up	up(s)

[router1]

router2:

[router2]display ip interface brief

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

The number of interface that is UP in Physical is 3

The number of interface that is DOWN in Physical is 1

The number of interface that is UP in Protocol is 1

The number of interface that is DOWN in Protocol is 3

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	up	down
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	unassigned	up	down
NULL0	unassigned	up	up(s)

[router2]

router3:

[router3]display ip interface brief

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

The number of interface that is UP in Physical is 3

The number of interface that is DOWN in Physical is 1

The number of interface that is UP in Protocol is 1

The number of interface that is DOWN in Protocol is 3

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	unassigned	up	down
GigabitEthernet0/0/2	unassigned	up	down
NULL0	unassigned	up	up(s)

[router3]

Таблицы маршрутизации:

router1:

[router1]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 4

Routes : 4

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
------------------	-------	-----	------	-------	---------	-----------

127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[router1]

router2:

[router2]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 4

Routes : 4

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[router2]

router3:

[router3]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 4

Routes : 4

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[router3]

1.3 Настройка IP-адреса для физических интерфейсов

router1:

[router1]interface GigabitEthernet0/0/1

[router1-GigabitEthernet0/0/1]ip address 10.0.13.1 24

Oct 9 2025 05:44:00-08:00 router1 %%01IFNET/4/LINK_STATE(l)[0]:The line
protoco

l IP on the interface GigabitEthernet0/0/1 has entered the UP state.

[router1-GigabitEthernet0/0/1]

[router1-GigabitEthernet0/0/1]quit

[router1]

```
[router1]interface GigabitEthernet0/0/0
[router1-GigabitEthernet0/0/0]ip address 10.0.12.1 24
Oct  9 2025 05:45:22-08:00 router1 %%01IFNET/4/LINK_STATE(l)[1]:The line
protoco
l IP on the interface GigabitEthernet0/0/0 has entered the UP state.
[router1-GigabitEthernet0/0/0]quit
[router1]
```

router2:

```
[router2]interface GigabitEthernet0/0/0
[router2-GigabitEthernet0/0/0]ip address 10.0.12.2 24
[router2-GigabitEthernet0/0/0]
Oct  9 2025 05:46:37-08:00 router2 %%01IFNET/4/LINK_STATE(l)[0]:The line
protoco
l IP on the interface GigabitEthernet0/0/0 has entered the UP state.
[router2-GigabitEthernet0/0/0]quit
[router2]
```

```
[router2]interface GigabitEthernet0/0/2
[router2-GigabitEthernet0/0/2]ip address 10.0.23.2 24
[router2-GigabitEthernet0/0/2]
Oct  9 2025 05:47:43-08:00 router2 %%01IFNET/4/LINK_STATE(l)[1]:The line
protoco
l IP on the interface GigabitEthernet0/0/2 has entered the UP state.
[router2-GigabitEthernet0/0/2]quit
[router2]
```

router3:

```
[router3]interface GigabitEthernet0/0/1
[router3-GigabitEthernet0/0/1]ip address 10.0.13.3 24
Oct  9 2025 05:49:12-08:00 router3 %%01IFNET/4/LINK_STATE(l)[0]:The line
protoco
l IP on the interface GigabitEthernet0/0/1 has entered the UP state.
[router3-GigabitEthernet0/0/1]quit
```

```
[router3]interface GigabitEthernet0/0/2
[router3-GigabitEthernet0/0/2]ip address 10.0.23.3 24
Oct  9 2025 05:49:49-08:00 router3 %%01IFNET/4/LINK_STATE(l)[1]:The line
protoco
l IP on the interface GigabitEthernet0/0/2 has entered the UP state.
[router3-GigabitEthernet0/0/2]quit
[router3]
```

Проверка наличия связи:

router1:

```
[router1]ping 10.0.12.2
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=790 ms
  Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=70 ms
  Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=40 ms
  Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=50 ms

--- 10.0.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/196/790 ms
```

[router1]

```
[router1]ping 10.0.13.3
PING 10.0.13.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=480 ms
  Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=60 ms
  Reply from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 10.0.13.3: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.13.3: bytes=56 Sequence=5 ttl=255 time=40 ms

--- 10.0.13.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/130/480 ms
```

[router1]

router2:

```
<router2>ping 10.0.12.1
PING 10.0.12.1: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=255 time=140 ms
  Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=255 time=50 ms
  Reply from 10.0.12.1: bytes=56 Sequence=3 ttl=255 time=40 ms
  Reply from 10.0.12.1: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.12.1: bytes=56 Sequence=5 ttl=255 time=40 ms

--- 10.0.12.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/62/140 ms
```

<router2>

```
<router2>ping 10.0.23.3
```



```
PING 10.0.23.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=260 ms
  Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=50 ms
  Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=50 ms
  Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=255 time=60 ms
```

```
--- 10.0.23.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/92/260 ms
```

<router2>

router3:

```
<router3>ping 10.0.13.1
PING 10.0.13.1: 56 data bytes, press CTRL_C to break
  Reply from 10.0.13.1: bytes=56 Sequence=1 ttl=255 time=140 ms
  Reply from 10.0.13.1: bytes=56 Sequence=2 ttl=255 time=50 ms
  Reply from 10.0.13.1: bytes=56 Sequence=3 ttl=255 time=40 ms
  Reply from 10.0.13.1: bytes=56 Sequence=4 ttl=255 time=20 ms
  Reply from 10.0.13.1: bytes=56 Sequence=5 ttl=255 time=40 ms
```

```
--- 10.0.13.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/58/140 ms
```

<router3>

```
<router3>ping 10.0.23.2
PING 10.0.23.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.23.2: bytes=56 Sequence=1 ttl=255 time=50 ms
  Reply from 10.0.23.2: bytes=56 Sequence=2 ttl=255 time=50 ms
  Reply from 10.0.23.2: bytes=56 Sequence=3 ttl=255 time=40 ms
  Reply from 10.0.23.2: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.23.2: bytes=56 Sequence=5 ttl=255 time=40 ms
```

```
--- 10.0.23.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/44/50 ms
```

<router3>

Таблицы маршрутизации после конфигурации:

router1:

```
<router1>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----  
----
```

```
Routing Tables: Public
```

```
Destinations : 10
```

```
Routes : 10
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet
0/0/0						
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet
0/0/1						
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
<router1>
```

router2:

```
<router2>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----  
----
```

```
Routing Tables: Public
```

```
Destinations : 10
```

```
Routes : 10
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet
0/0/0						
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet
0/0/2						
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						

127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

router3:

```

<router3>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
----
Routing Tables: Public
    Destinations : 10          Routes : 10

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
10.0.13.0/24 Direct  0    0             D  10.0.13.3      GigabitEthernet
0/0/1
10.0.13.3/32 Direct  0    0             D  127.0.0.1      GigabitEthernet
0/0/1
10.0.13.255/32 Direct  0    0             D  127.0.0.1      GigabitEthernet
0/0/1
10.0.23.0/24 Direct  0    0             D  10.0.23.3      GigabitEthernet
0/0/2
10.0.23.3/32 Direct  0    0             D  127.0.0.1      GigabitEthernet
0/0/2
10.0.23.255/32 Direct  0    0             D  127.0.0.1      GigabitEthernet
0/0/2
127.0.0.0/8 Direct  0    0             D  127.0.0.1      InLoopBack0
127.0.0.1/32 Direct  0    0             D  127.0.0.1      InLoopBack0
127.255.255.255/32 Direct  0    0             D  127.0.0.1      InLoopBack0
255.255.255.255/32 Direct  0    0             D  127.0.0.1      InLoopBack0

<router3>

```

1.4 Создание loopback-интерфейса

router1:

```

[Huawei]interface LoopBack0
[Huawei-LoopBack0]ip address 10.0.1.1 32
[Huawei-LoopBack0]

```

router2:

```

[Huawei]interface LoopBack0
[Huawei-LoopBack0]ip address 10.0.1.2 32
[Huawei-LoopBack0]

```

router3:

```
[Huawei]interface LoopBack0
[Huawei-LoopBack0]ip address 10.0.1.3 32
[Huawei-LoopBack0]
```

Таблицы маршрутизации:

```
<router1>system-view
Enter system view, return user view with Ctrl+Z.
[router1]display ip routing-table
Route Flags: R - relay, D - download to fib
```

```
-----
----
Routing Tables: Public
```

```
Destinations : 11
```

```
Routes : 11
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet
0/0/0 10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0 10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0 10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet
0/0/1 10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1 10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1 127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
[router1]
```

```
[router2]display ip routing-table
Route Flags: R - relay, D - download to fib
```

```
-----
----
Routing Tables: Public
```

```
Destinations : 11
```

```
Routes : 11
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet
0/0/0 10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0 10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet

```

0/0/0
  10.0.23.0/24 Direct 0 0 D 10.0.23.2 GigabitEthernet
0/0/2
  10.0.23.2/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/2
  10.0.23.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/2
  127.0.0.0/8 Direct 0 0 D 127.0.0.1 InLoopBack0
  127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

```

[router2]

[router3]display ip routing-table

Route Flags: R - relay, D - download to fib

```

-----
----
Routing Tables: Public
      Destinations : 11          Routes : 11

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
      10.0.1.3/32    Direct  0    0             D  127.0.0.1          LoopBack0
0/0/1  10.0.13.0/24      Direct  0    0            D  10.0.13.3          GigabitEthernet
0/0/1  10.0.13.3/32      Direct  0    0            D  127.0.0.1          GigabitEthernet
0/0/1  10.0.13.255/32    Direct  0    0            D  127.0.0.1          GigabitEthernet
0/0/1  10.0.23.0/24      Direct  0    0            D  10.0.23.3          GigabitEthernet
0/0/2  10.0.23.3/32      Direct  0    0            D  127.0.0.1          GigabitEthernet
0/0/2  10.0.23.255/32    Direct  0    0            D  127.0.0.1          GigabitEthernet
0/0/2  127.0.0.0/8       Direct  0    0            D  127.0.0.1          InLoopBack0
      127.0.0.1/32    Direct  0    0            D  127.0.0.1          InLoopBack0
127.255.255.255/32  Direct  0    0            D  127.0.0.1          InLoopBack0
255.255.255.255/32  Direct  0    0            D  127.0.0.1          InLoopBack0

```

[router3]

Отсутствие связи на loopback без статических маршрутов:

[router1]ping -a 10.0.1.1 10.0.1.2

PING 10.0.1.2: 56 data bytes, press CTRL_C to break

```

Request time out
Request time out
Request time out
Request time out
Request time out

```

```
--- 10.0.1.2 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

[router1]

```
[router2]ping -a 10.0.1.2 10.0.1.3
PING 10.0.1.3: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out
```

```
--- 10.0.1.3 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

[router2]

```
[router3]ping -a 10.0.1.3 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out
```

```
--- 10.0.1.1 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

[router3]

1.5 Настройка статического маршрута

Настроим один статический маршрут:

```
[router1]ip route-static 10.0.1.2 32 10.0.12.2
[router1]ip route-static 10.0.1.3 32 10.0.13.3
[router1]display ip routing-table
Route Flags: R - relay, D - download to fib
```

```
-----
Routing Tables: Public
Destinations : 13          Routes : 13
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.2/32	Static	60	0	RD	10.0.12.2	GigabitEthernet
0/0/0						
10.0.1.3/32	Static	60	0	RD	10.0.13.3	GigabitEthernet
0/0/1						
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet
0/0/0						
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet
0/0/1						
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
[router1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.1.2 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

```
[router2]ip route-static 10.0.1.1 32 10.0.12.1
```

Наличие связи по loopback на router1:

```
[router1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=255 time=180 ms
Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=255 time=50 ms
Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=255 time=30 ms
Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=255 time=50 ms
Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.1.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
```

0.00% packet loss
round-trip min/avg/max = 30/68/180 ms

[router1]

Установка оставшихся статических маршрутов:

```
[router2]ip route-static 10.0.1.3 32 10.0.23.3  
[router3]ip route-static 10.0.1.1 32 10.0.13.1  
[router3]ip route-static 10.0.1.2 32 10.0.23.2
```

Проверка остальных связей:

```
[router1]ping -a 10.0.1.1 10.0.1.3  
PING 10.0.1.3: 56 data bytes, press CTRL_C to break  
  Reply from 10.0.1.3: bytes=56 Sequence=1 ttl=255 time=900 ms  
  Reply from 10.0.1.3: bytes=56 Sequence=2 ttl=255 time=750 ms  
  Reply from 10.0.1.3: bytes=56 Sequence=3 ttl=255 time=760 ms  
  Reply from 10.0.1.3: bytes=56 Sequence=4 ttl=255 time=180 ms  
  Reply from 10.0.1.3: bytes=56 Sequence=5 ttl=255 time=40 ms  
  
--- 10.0.1.3 ping statistics ---  
  5 packet(s) transmitted  
  5 packet(s) received  
  0.00% packet loss  
  round-trip min/avg/max = 40/526/900 ms
```

[router1]

```
[router2]ping -a 10.0.1.2 10.0.1.1  
PING 10.0.1.1: 56 data bytes, press CTRL_C to break  
  Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=255 time=80 ms  
  Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=255 time=80 ms  
  Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=255 time=70 ms  
  Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=255 time=60 ms  
  Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=255 time=110 ms  
  
--- 10.0.1.1 ping statistics ---  
  5 packet(s) transmitted  
  5 packet(s) received  
  0.00% packet loss  
  round-trip min/avg/max = 60/80/110 ms
```

```
[router2]ping -a 10.0.1.2 10.0.1.3  
PING 10.0.1.3: 56 data bytes, press CTRL_C to break  
  Reply from 10.0.1.3: bytes=56 Sequence=1 ttl=255 time=100 ms  
  Reply from 10.0.1.3: bytes=56 Sequence=2 ttl=255 time=20 ms  
  Reply from 10.0.1.3: bytes=56 Sequence=3 ttl=255 time=120 ms  
  Reply from 10.0.1.3: bytes=56 Sequence=4 ttl=255 time=80 ms  
  Reply from 10.0.1.3: bytes=56 Sequence=5 ttl=255 time=70 ms
```



```
--- 10.0.1.3 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 20/78/120 ms
```

[router2]

```
[router3]ping -a 10.0.1.3 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
  Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=255 time=110 ms
  Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=255 time=20 ms
  Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=255 time=50 ms

--- 10.0.1.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 20/50/110 ms
```

```
[router3]ping -a 10.0.1.3 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=255 time=40 ms
  Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=255 time=70 ms
  Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.1.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 30/42/70 ms
```

[router3]

1.6 Настройка маршрута от R₁ к R₂ через R₃ в качестве резервного маршрута от LoopBacko R₁ к LoopBacko R₂

Маршрут между r1 и r2 через r3:

```
[router1]ip route-static 10.0.1.2 32 10.0.13.3 preference 100
[router2]ip route-static 10.0.1.1 32 10.0.23.3 preference 100
```

```
[router1]display ip routing-table
Route Flags: R - relay, D - download to fib
```

```

-----
----
Routing Tables: Public
      Destinations : 13          Routes : 13

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
    10.0.1.1/32    Direct   0     0           D    127.0.0.1        LoopBack0
    10.0.1.2/32    Static   60    0           RD   10.0.12.2        GigabitEthernet
0/0/0
    10.0.1.3/32    Static   60    0           RD   10.0.13.3        GigabitEthernet
0/0/1
    10.0.12.0/24   Direct   0     0           D    10.0.12.1        GigabitEthernet
0/0/0
    10.0.12.1/32   Direct   0     0           D    127.0.0.1        GigabitEthernet
0/0/0
    10.0.12.255/32 Direct   0     0           D    127.0.0.1        GigabitEthernet
0/0/0
    10.0.13.0/24   Direct   0     0           D    10.0.13.1        GigabitEthernet
0/0/1
    10.0.13.1/32   Direct   0     0           D    127.0.0.1        GigabitEthernet
0/0/1
    10.0.13.255/32 Direct   0     0           D    127.0.0.1        GigabitEthernet
0/0/1
    127.0.0.0/8    Direct   0     0           D    127.0.0.1        InLoopBack0
    127.0.0.1/32   Direct   0     0           D    127.0.0.1        InLoopBack0
127.255.255.255/32 Direct   0     0           D    127.0.0.1        InLoopBack0
255.255.255.255/32 Direct   0     0           D    127.0.0.1        InLoopBack0

```

[router1]

```

[router2]display ip routing-table
Route Flags: R - relay, D - download to fib

```

```

-----
----
Routing Tables: Public
      Destinations : 13          Routes : 13

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
    10.0.1.1/32    Static   60    0           RD   10.0.12.1        GigabitEthernet
0/0/0
    10.0.1.2/32    Direct   0     0           D    127.0.0.1        LoopBack0
    10.0.1.3/32    Static   60    0           RD   10.0.23.3        GigabitEthernet
0/0/2
    10.0.12.0/24   Direct   0     0           D    10.0.12.2        GigabitEthernet
0/0/0
    10.0.12.2/32   Direct   0     0           D    127.0.0.1        GigabitEthernet
0/0/0
    10.0.12.255/32 Direct   0     0           D    127.0.0.1        GigabitEthernet
0/0/0
    10.0.23.0/24   Direct   0     0           D    10.0.23.2        GigabitEthernet
0/0/2

```

```

10.0.23.2/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/2
10.0.23.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/2
127.0.0.0/8 Direct 0 0 D 127.0.0.1 InLoopBack0
127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

```

[router2]

Отключение интерфейсов между r1 и r2:

```

[router1]interface GigabitEthernet0/0/0
[router1-GigabitEthernet0/0/0]shutdown
[router1-GigabitEthernet0/0/0]
Oct 9 2025 16:57:17-08:00 router1 %%01IFPDT/4/IF_STATE(l)[3]:Interface
GigabitE
thernet0/0/0 has turned into DOWN state.
Oct 9 2025 16:57:17-08:00 router1 %%01IFNET/4/LINK_STATE(l)[4]:The line
protoco
l IP on the interface GigabitEthernet0/0/0 has entered the DOWN state.
[router1-GigabitEthernet0/0/0]

```

```

[router2]interface GigabitEthernet0/0/0
[router2-GigabitEthernet0/0/0]shutdown
Oct 9 2025 16:57:56-08:00 router2 %%01IFPDT/4/IF_STATE(l)[6]:Interface
GigabitE
thernet0/0/0 has turned into DOWN state.
[router2-GigabitEthernet0/0/0]
[router2-GigabitEthernet0/0/0]

```

Демонстрация активации маршрутов с низким приоритетом при отключении узла с высоким:

```

[router1]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
----
Routing Tables: Public
Destinations : 10 Routes : 10

Destination/Mask    Proto    Pre    Cost    Flags NextHop          Interface
10.0.1.1/32        Direct   0       0         D  127.0.0.1          LoopBack0
10.0.1.2/32        Static  100     0         RD  10.0.13.3          GigabitEthernet
0/0/1
10.0.1.3/32        Static   60      0         RD  10.0.13.3          GigabitEthernet
0/0/1
10.0.13.0/24       Direct   0       0         D  10.0.13.1          GigabitEthernet

```

```

0/0/1
  10.0.13.1/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/1
  10.0.13.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/1
  127.0.0.0/8 Direct 0 0 D 127.0.0.1 InLoopBack0
  127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

```

[router1]

```

[router2]display IP routing-table
Route Flags: R - relay, D - download to fib

```


 Routing Tables: Public

Destinations : 10 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0/0/2 10.0.1.1/32	Static	100	0	RD	10.0.23.3	GigabitEthernet
10.0.1.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
0/0/2 10.0.1.3/32	Static	60	0	RD	10.0.23.3	GigabitEthernet
0/0/2 10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet
0/0/2 10.0.23.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2 10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2 127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[router2]

Доступность узла по резервному маршруту:

```

[router1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=254 time=240 ms
  Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=254 time=80 ms
  Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=254 time=50 ms
  Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=254 time=50 ms
  Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=254 time=80 ms

--- 10.0.1.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received

```

0.00% packet loss
round-trip min/avg/max = 50/100/240 ms

[router1]

[router2]ping -a 10.0.1.2 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=254 time=40 ms
Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=254 time=50 ms
Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=254 time=60 ms
Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=254 time=60 ms
Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=254 time=50 ms

--- 10.0.1.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 40/52/60 ms

[router2]

Трассировки маршрутов:

[router1]tracert -a 10.0.1.1 10.0.1.2

tracert to 10.0.1.2(10.0.1.2), max hops: 30 ,packet length: 40,press
CTRL_C
to break

1 10.0.13.3 780 ms 130 ms 40 ms

2 10.0.23.2 110 ms 80 ms 40 ms
[router1]

[router2]tracert -a 10.0.1.2 10.0.1.1

tracert to 10.0.1.1(10.0.1.1), max hops: 30 ,packet length: 40,press
CTRL_C
to break

1 10.0.23.3 60 ms 30 ms 40 ms

2 10.0.13.1 140 ms 70 ms 70 ms
[router2]

1.7 Настройка маршрута по умолчанию для установления связи между интерфейсом LoopBack0 маршрутизатора R₁ и интерфейсом LoopBack0 маршрутизатора R₂

Удаление маршрутов:

```
[router1-GigabitEthernet0/0/0]undo shutdown
[router1-GigabitEthernet0/0/0]quit
[router1]undo ip route-static 10.0.1.2 255.255.255.255 10.0.12.2
[router1]undo ip route-static 10.0.1.2 255.255.255.255 10.0.13.3 preference 100
```

Отсутствие маршрута на router2:

```
[router1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
----
Routing Tables: Public
      Destinations : 9          Routes : 9

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
-----
10.0.1.1/32        Direct   0    0                D    127.0.0.1          LoopBack0
10.0.1.3/32        Static   60    0               RD    10.0.13.3          GigabitEthernet
0/0/1
10.0.13.0/24       Direct   0    0                D    10.0.13.1          GigabitEthernet
0/0/1
10.0.13.1/32       Direct   0    0                D    127.0.0.1          GigabitEthernet
0/0/1
10.0.13.255/32     Direct   0    0                D    127.0.0.1          GigabitEthernet
0/0/1
127.0.0.0/8        Direct   0    0                D    127.0.0.1          InLoopBack0
127.0.0.1/32       Direct   0    0                D    127.0.0.1          InLoopBack0
127.255.255.255/32 Direct   0    0                D    127.0.0.1          InLoopBack0
255.255.255.255/32 Direct   0    0                D    127.0.0.1          InLoopBack0

[router1]
```

Маршрут по умолчанию на router1:

```
[router1]ip route-static 0.0.0.0 0 10.0.12.2
[router1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
----
Routing Tables: Public
      Destinations : 9          Routes : 9

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
```

	10.0.1.1/32	Direct	0	0		D	127.0.0.1	LoopBack0
0/0/1	10.0.1.3/32	Static	60	0	RD	10.0.13.3		GigabitEthernet
	10.0.13.0/24	Direct	0	0		D	10.0.13.1	GigabitEthernet
0/0/1	10.0.13.1/32	Direct	0	0		D	127.0.0.1	GigabitEthernet
0/0/1	10.0.13.255/32	Direct	0	0		D	127.0.0.1	GigabitEthernet
0/0/1	127.0.0.0/8	Direct	0	0		D	127.0.0.1	InLoopBack0
	127.0.0.1/32	Direct	0	0		D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0		D	127.0.0.1	InLoopBack0	
255.255.255.255/32	Direct	0	0		D	127.0.0.1	InLoopBack0	

[router1]

Наличие путей между router1 и router2:

```
[router1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=255 time=60 ms
  Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=255 time=40 ms
  Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=255 time=50 ms
  Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=255 time=50 ms

--- 10.0.1.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/48/60 ms
```

[router1]

Итоговые конфигурации:

router1:

```
<router1>display current-configuration
[V200R003C00]
#
 sysname router1
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load portalpage.zip
#
 drop illegal-mac alarm
```

```

#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
ip address 10.0.12.1 255.255.255.0
#
interface GigabitEthernet0/0/1
ip address 10.0.13.1 255.255.255.0
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
interface LoopBack0
ip address 10.0.1.1 255.255.255.255
#
ip route-static 0.0.0.0 0.0.0.0 10.0.12.2
ip route-static 10.0.1.3 255.255.255.255 10.0.13.3
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
<router1>

```

router2:

```

<router2>dis cu
[V200R003C00]
#
sysname router2
#
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#

```



```

portal local-server load portalpage.zip
#
drop illegal-mac alarm
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
ip address 10.0.12.2 255.255.255.0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
ip address 10.0.23.2 255.255.255.0
#
interface NULL0
#
interface LoopBack0
ip address 10.0.1.2 255.255.255.255
#
ip route-static 10.0.1.1 255.255.255.255 10.0.12.1
ip route-static 10.0.1.1 255.255.255.255 10.0.23.3 preference 100
ip route-static 10.0.1.3 255.255.255.255 10.0.23.3
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
<router2>

```

router3:

```

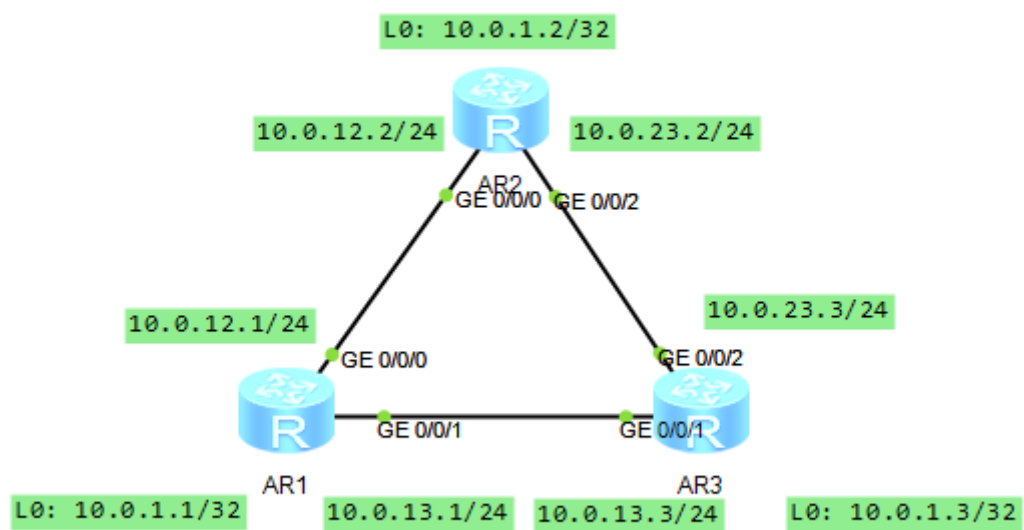
<router3>dis cu
[V200R003C00]
#
sysname router3
#
snmp-agent local-engineid 800007DB0300000000000000

```

```

snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load portalpage.zip
#
drop illegal-mac alarm
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
ip address 10.0.13.3 255.255.255.0
#
interface GigabitEthernet0/0/2
ip address 10.0.23.3 255.255.255.0
#
interface NULL0
#
interface LoopBack0
ip address 10.0.1.3 255.255.255.255
#
ip route-static 10.0.1.1 255.255.255.255 10.0.13.1
ip route-static 10.0.1.2 255.255.255.255 10.0.23.2
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
<router3>

```



2 Маршрутизация OSPF

2.1 Настройка основных параметров устройств

router1:

```
[router1]display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----
```

Routing Tables: Public
Destinations : 12 Routes : 12

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
0/0/0 10.0.1.2/32	Static	60	0	RD	10.0.12.2	GigabitEthernet
0/0/0 10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet
0/0/0 10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0 10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0 10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet
0/0/1 10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1 10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1 127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
[router1]
```

router2:

```
<router2>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----
```

Routing Tables: Public
Destinations : 11 Routes : 11

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
0/0/0 10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet
0/0/0 10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0 10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0 10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet

```

0/0/2
    10.0.23.2/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/2
    10.0.23.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/2
    127.0.0.0/8 Direct 0 0 D 127.0.0.1 InLoopBack0
    127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

```

<router2>

router3:

```

[router3]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
----
Routing Tables: Public
      Destinations : 13          Routes : 13

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
    10.0.1.1/32 Static  60   0           RD  10.0.13.1          GigabitEthernet
0/0/1
    10.0.1.2/32 Static  60   0           RD  10.0.23.2          GigabitEthernet
0/0/2
    10.0.1.3/32 Direct   0    0           D   127.0.0.1          LoopBack0
    10.0.13.0/24 Direct   0    0           D   10.0.13.3          GigabitEthernet
0/0/1
    10.0.13.3/32 Direct   0    0           D   127.0.0.1          GigabitEthernet
0/0/1
    10.0.13.255/32 Direct   0    0           D   127.0.0.1          GigabitEthernet
0/0/1
    10.0.23.0/24 Direct   0    0           D   10.0.23.3          GigabitEthernet
0/0/2
    10.0.23.3/32 Direct   0    0           D   127.0.0.1          GigabitEthernet
0/0/2
    10.0.23.255/32 Direct   0    0           D   127.0.0.1          GigabitEthernet
0/0/2
    127.0.0.0/8 Direct   0    0           D   127.0.0.1          InLoopBack0
    127.0.0.1/32 Direct   0    0           D   127.0.0.1          InLoopBack0
127.255.255.255/32 Direct   0    0           D   127.0.0.1          InLoopBack0
255.255.255.255/32 Direct   0    0           D   127.0.0.1          InLoopBack0

```

[router3]

2.2 Настройка основных параметров OSPF

Router1:

```
[router1]ospf 1
[router1-ospf-1]area 0
[router1-ospf-1-area-0.0.0.0]network 10.0.12.1 0.0.0.255
[router1-ospf-1-area-0.0.0.0]network 10.0.13.1 0.0.0.255
[router1-ospf-1-area-0.0.0.0]network 10.0.1.1 0.0.0.0
[router1-ospf-1-area-0.0.0.0]
```

Router2:

```
[router2]ospf 1
[router2-ospf-1]area 0
[router2-ospf-1-area-0.0.0.0]network 10.0.12.1 0.0.0.255
[router2-ospf-1-area-0.0.0.0]network 10.0.13.1 0.0.0.255
[router2-ospf-1-area-0.0.0.0]network 10.0.1.1 0.0.0.0
[router2-ospf-1-area-0.0.0.0]
```

Router3:

```
[router3]ospf 1
[router3-ospf-1]area 0
[router3-ospf-1-area-0.0.0.0]network 10.0.12.1 0.0.0.255
[router3-ospf-1-area-0.0.0.0]network 10.0.13.1 0.0.0.255
[router3-ospf-1-area-0.0.0.0]network 10.0.1.1 0.0.0.0
[router3-ospf-1-area-0.0.0.0]
```

2.3 Вывод на экран рабочих статусов OSPF

Вывод информации о соседях OSPF:

```
[router1]display ospf peer
```

```
OSPF Process 1 with Router ID 10.0.12.1
Neighbors
```

```
Area 0.0.0.0 interface 10.0.12.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.12.2      Address: 10.0.12.2
State: Full  Mode:Nbr is Master  Priority: 1
DR: 10.0.12.1  BDR: 10.0.12.2  MTU: 0
Dead timer due in 30  sec
Retrans timer interval: 5
Neighbor is up for 00:02:37
Authentication Sequence: [ 0 ]
```

Neighbors

```
Area 0.0.0.0 interface 10.0.13.1(GigabitEthernet0/0/1)'s neighbors
Router ID: 10.0.13.3      Address: 10.0.13.3
State: Full  Mode:Nbr is Master  Priority: 1
DR: 10.0.13.1  BDR: 10.0.13.3  MTU: 0
Dead timer due in 35  sec
```

```
Retrans timer interval: 5
Neighbor is up for 00:01:56
Authentication Sequence: [ 0 ]
```

[router1]

[router2]display ospf peer

```
OSPF Process 1 with Router ID 10.0.12.2
Neighbors
```

```
Area 0.0.0.0 interface 10.0.12.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.12.1      Address: 10.0.12.1
State: Full  Mode:Nbr is Slave  Priority: 1
DR: 10.0.12.1  BDR: 10.0.12.2  MTU: 0
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00:03:15
Authentication Sequence: [ 0 ]
```

Neighbors

```
Area 0.0.0.0 interface 10.0.23.2(GigabitEthernet0/0/2)'s neighbors
Router ID: 10.0.13.3      Address: 10.0.23.3
State: Full  Mode:Nbr is Master Priority: 1
DR: 10.0.23.2  BDR: 10.0.23.3  MTU: 0
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00:02:32
Authentication Sequence: [ 0 ]
```

[router2]

router3]display ospf peer

```
OSPF Process 1 with Router ID 10.0.13.3
Neighbors
```

```
Area 0.0.0.0 interface 10.0.13.3(GigabitEthernet0/0/1)'s neighbors
Router ID: 10.0.12.1      Address: 10.0.13.1
State: Full  Mode:Nbr is Slave  Priority: 1
DR: 10.0.13.1  BDR: 10.0.13.3  MTU: 0
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00:03:02
Authentication Sequence: [ 0 ]
```

Neighbors

```
Area 0.0.0.0 interface 10.0.23.3(GigabitEthernet0/0/2)'s neighbors
Router ID: 10.0.12.2      Address: 10.0.23.2
State: Full  Mode:Nbr is Slave  Priority: 1
```

DR: 10.0.23.2 BDR: 10.0.23.3 MTU: 0
Dead timer due in 35 sec
Retrans timer interval: 0
Neighbor is up for 00:03:00
Authentication Sequence: [0]

Маршруты полученные от OSPF:

```
[router1]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib
```

```
Public routing table : OSPF
```

```
Destinations : 3 Routes : 4
```

```
OSPF routing table status : <Active>
```

```
Destinations : 3 Routes : 4
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0/0/0 10.0.1.2/32	OSPF	10	1	D	10.0.12.2	GigabitEthernet
0/0/1 10.0.1.3/32	OSPF	10	1	D	10.0.13.3	GigabitEthernet
0/0/0 10.0.23.0/24	OSPF	10	2	D	10.0.12.2	GigabitEthernet
0/0/1	OSPF	10	2	D	10.0.13.3	GigabitEthernet

```
OSPF routing table status : <Inactive>
```

```
Destinations : 0 Routes : 0
```

```
[router2]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib
```

```
Public routing table : OSPF
```

```
Destinations : 3 Routes : 4
```

```
OSPF routing table status : <Active>
```

```
Destinations : 3 Routes : 4
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0/0/0 10.0.1.1/32	OSPF	10	1	D	10.0.12.1	GigabitEthernet
0/0/2 10.0.1.3/32	OSPF	10	1	D	10.0.23.3	GigabitEthernet
0/0/0 10.0.13.0/24	OSPF	10	2	D	10.0.12.1	GigabitEthernet
0/0/2	OSPF	10	2	D	10.0.23.3	GigabitEthernet


```
OSPF routing table status : <Inactive>
Destinations : 0          Routes : 0
```

```
[router2]
```

```
[router3]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib
```

```
-----
Public routing table : OSPF
Destinations : 3          Routes : 4
```

```
OSPF routing table status : <Active>
Destinations : 3          Routes : 4
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0/0/1 10.0.1.1/32	OSPF	10	1	D	10.0.13.1	GigabitEthernet
0/0/2 10.0.1.2/32	OSPF	10	1	D	10.0.23.2	GigabitEthernet
0/0/1 10.0.12.0/24	OSPF	10	2	D	10.0.13.1	GigabitEthernet
0/0/2	OSPF	10	2	D	10.0.23.2	GigabitEthernet

```
OSPF routing table status : <Inactive>
Destinations : 0          Routes : 0
```

```
[router3]
```

2.4 Настройка аутентификации OSPF

Настройка аутентификации OSPF для router1:

```
[router1]interface GigabitEthernet0/0/1
[router1-GigabitEthernet0/0/1]ospf authentication-mode md5 1 cipher HCIA-Datacom
```

```
[router1]interface GigabitEthernet0/0/0
[router1-GigabitEthernet0/0/0]
Oct 9 2025 20:48:43-08:00 router1 %%010SPF/3/NBR_CHG_DOWN(1)[11]:Neighbor
event
:neighbor state changed to Down. (ProcessId=256, NeighborAddress=3.13.0.10,
Neig
hborEvent=InactivityTimer, NeighborPreviousState=Full,
NeighborCurrentState=Down
)
[router1-GigabitEthernet0/0/0]
```

```

Oct          9          2025          20:48:43-08:00          router1
%%010SPF/3/NBR_DOWN_REASON(l)[12]:Neighbor st
ate leaves full or changed to Down. (ProcessId=256,
NeighborRouterId=3.13.0.10,
NeighborAreaId=0,
NeighborInterface=GigabitEthernet0/0/1,NeighborDownImmediate r
eason=Neighbor Down Due to Inactivity, NeighborDownPrimeReason=Hello Not
Seen, N
eighborChangeTime=2025-10-09 20:48:43-08:00)
[router1-GigabitEthernet0/0/0]ospf authentication-mode md5 1 cipher HCIA-
Datacom

[router1-GigabitEthernet0/0/0]display this
[V200R003C00]
#
interface GigabitEthernet0/0/0
 ip address 10.0.12.1 255.255.255.0
 ospf authentication-mode md5 1 cipher %$$$+BZ)(nXVA)'{:b<[RbY,1"LY%$$$
#
return
[router1-GigabitEthernet0/0/0]

```

Вывод соседей OSPF для router1:

```

[router1]display ospf peer brief

      OSPF Process 1 with Router ID 10.0.12.1
      Peer Statistic Information
-----
Area Id          Interface          Neighbor id          State
-----
---
[router1]

```

Для router2:

```

[router2]
Oct          9          2025          20:51:55-08:00          router2
%%010SPF/3/NBR_DOWN_REASON(l)[21]:Neighbor st
ate leaves full or changed to Down. (ProcessId=256,
NeighborRouterId=3.13.0.10,
NeighborAreaId=0,
NeighborInterface=GigabitEthernet0/0/2,NeighborDownImmediate r
eason=Neighbor Down Due to Inactivity, NeighborDownPrimeReason=Hello Not
Seen, N
eighborChangeTime=2025-10-09 20:51:55-08:00)
[router2]display ospf peer brief

      OSPF Process 1 with Router ID 10.0.12.2
      Peer Statistic Information

```

```

-----
Area Id          Interface          Neighbor id      State
0.0.0.0          GigabitEthernet0/0/0  10.0.12.1       Full
-----
[router2]

```

Для router3:

```

[router3]display ospf peer brief

  OSPF Process 1 with Router ID 10.0.13.3
    Peer Statistic Information
-----
Area Id          Interface          Neighbor id      State
0.0.0.0          GigabitEthernet0/0/1  10.0.12.1       Full
0.0.0.0          GigabitEthernet0/0/2  10.0.12.2       Full
-----
[router3]

```

2.5 Установка маршрута по умолчанию

Таблицы маршрутизации для router2 и router3:

```

[router2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
----
Routing Tables: Public
      Destinations : 15          Routes : 16

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
0.0.0.0/0           0_ASE    150  1              D    10.0.12.1          GigabitEthernet
10.0.1.1/32         OSPF     10   1              D    10.0.12.1          GigabitEthernet
10.0.1.2/32         Direct   0     0              D    127.0.0.1          LoopBack0
10.0.1.3/32         OSPF     10   1              D    10.0.23.3          GigabitEthernet
10.0.12.0/24        Direct   0     0              D    10.0.12.2          GigabitEthernet
10.0.12.2/32        Direct   0     0              D    127.0.0.1          GigabitEthernet
10.0.12.255/32      Direct   0     0              D    127.0.0.1          GigabitEthernet
10.0.13.0/24        OSPF     10   2              D    10.0.12.1          GigabitEthernet

```

Destination/Interface	Proto	Pre	Cost	Flags	NextHop	Interface	
0/0/2	OSPF	10	2	D	10.0.23.3	GigabitEthernet	
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet	
0/0/2	10.0.23.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2	10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0	
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0	
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0	
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0	

[router2]

[router3]display ip routing-table

Route Flags: R - relay, D - download to fib

```

-----
----
Routing Tables: Public
    Destinations : 15          Routes : 16

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
0/0/1               0.0.0.0/0  O_ASE  150  1             D  10.0.13.1         GigabitEthernet
0/0/1               10.0.1.1/32 OSPF    10   1             D  10.0.13.1         GigabitEthernet
0/0/1               10.0.1.2/32 OSPF    10   1             D  10.0.23.2         GigabitEthernet
0/0/2               10.0.1.3/32 Direct   0    0             D  127.0.0.1         LoopBack0
0/0/1               10.0.12.0/24 OSPF    10   2             D  10.0.13.1         GigabitEthernet
0/0/2               OSPF    10   2             D  10.0.23.2         GigabitEthernet
0/0/2               10.0.13.0/24 Direct   0    0             D  10.0.13.3         GigabitEthernet
0/0/1               10.0.13.3/32 Direct   0    0             D  127.0.0.1         GigabitEthernet
0/0/1               10.0.13.255/32 Direct   0    0             D  127.0.0.1         GigabitEthernet
0/0/1               10.0.23.0/24 Direct   0    0             D  10.0.23.3         GigabitEthernet
0/0/2               10.0.23.3/32 Direct   0    0             D  127.0.0.1         GigabitEthernet
0/0/2               10.0.23.255/32 Direct   0    0             D  127.0.0.1         GigabitEthernet
0/0/2               127.0.0.0/8 Direct   0    0             D  127.0.0.1         InLoopBack0
0/0/2               127.0.0.1/32 Direct   0    0             D  127.0.0.1         InLoopBack0
127.255.255.255/32 Direct   0    0             D  127.0.0.1         InLoopBack0
255.255.255.255/32 Direct   0    0             D  127.0.0.1         InLoopBack0

```

[router3]

2.6 Изменения значений стоимости интерфейсов на R1, чтобы LoopBack0 на R1 мог достигать LoopBack0 на R2 через R3

```
[router1-ospf-1]quit
[router1]interface GigabitEthernet0/0/0
[router1-GigabitEthernet0/0/0]ospf cost 10
```

```
[router1]display ip routing-table
Route Flags: R - relay, D - download to fib
```

```
-----
----
Routing Tables: Public
```

```
Destinations : 14
```

```
Routes : 14
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.2/32	OSPF	10	2	D	10.0.13.3	GigabitEthernet0/0/1
10.0.1.3/32	OSPF	10	1	D	10.0.13.3	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet0/0/0
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/1
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.23.0/24	OSPF	10	2	D	10.0.13.3	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
[router1]
```

Проверим результат конфигурирования:

```
[router1]tracert -a 10.0.1.1 10.0.1.2
```

```
tracert to 10.0.1.2(10.0.1.2), max hops: 30 ,packet length: 40,press
CTRL_C
to break
```

```
1 10.0.13.3 420 ms 90 ms 60 ms
2 10.0.23.2 220 ms 70 ms 50 ms
[router1]
```

```
[router1]ping 10.0.12.2
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=170 ms
  Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=60 ms
  Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=60 ms
  Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=50 ms

--- 10.0.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/76/170 ms
```

```
[router3]ping 10.0.12.2
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=80 ms
  Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=40 ms
  Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=80 ms
  Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=50 ms
  Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=90 ms

--- 10.0.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/68/90 ms
```

Итоговые конфигурации устройств:

```
<router1>display current-configuration
[V200R003C00]
#
 sysname router1
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
```

```

set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
ip address 10.0.12.1 255.255.255.0
ospf cost 10
ospf authentication-mode md5 1 cipher %$$$+BZ)(nXVA)'{:b<[RbY,1"LY%$$$
#
interface GigabitEthernet0/0/1
ip address 10.0.13.1 255.255.255.0
ospf authentication-mode md5 1 cipher %$$$vSnLHnr{!@+$+`9%6t}^K1"Lg%$$$
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
interface LoopBack0
ip address 10.0.1.1 255.255.255.255
#
ospf 1
default-route-advertise always
area 0.0.0.0
network 10.0.1.1 0.0.0.0
network 10.0.12.0 0.0.0.255
network 10.0.13.0 0.0.0.255
#
ip route-static 10.0.1.2 255.255.255.255 10.0.12.2
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return

```

```
[V200R003C00]
```

```

#
sysname router2
#
snmp-agent local-engineid 800007DB03000000000000

```

```

snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load flash:/portalpage.zip
#
drop illegal-mac alarm
#
wlan ac-global carrier id other ac id 0
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
ip address 10.0.12.2 255.255.255.0
ospf authentication-mode md5 1 cipher %$$ob}|,~YD~7UJ^iDgcS3,1$*D%$$
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
ip address 10.0.23.2 255.255.255.0
ospf authentication-mode md5 1 cipher %$$!28z'|pqn%tcdeAeIV!J1%Bd%$$
#
interface NULL0
#
interface LoopBack0
ip address 10.0.1.2 255.255.255.255
#
ospf 1
area 0.0.0.0
network 10.0.1.2 0.0.0.0
network 10.0.12.2 0.0.0.0
network 10.0.23.2 0.0.0.0
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
<router2>

```



```

[V200R003C00]
#
 sysname router3
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
 set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$
 local-user admin service-type http
#
 firewall zone Local
 priority 15
#
 interface GigabitEthernet0/0/0
#
 interface GigabitEthernet0/0/1
 ip address 10.0.13.3 255.255.255.0
#
 interface GigabitEthernet0/0/2
 ip address 10.0.23.3 255.255.255.0
#
 interface NULL0
#
 interface LoopBack0
 ip address 10.0.1.3 255.255.255.255
#
 ospf 1
 area 0.0.0.0
 authentication-mode md5 1 cipher %$$-DhX&C3\b&0b3.<-J\nB1&0G%$$
 network 10.0.1.3 0.0.0.0
 network 10.0.13.3 0.0.0.0
 network 10.0.23.3 0.0.0.0
#
 ip route-static 10.0.1.1 255.255.255.255 10.0.13.1
 ip route-static 10.0.1.2 255.255.255.255 10.0.23.2
#
 user-interface con 0
 authentication-mode password

```

```
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
[router3]
```

ЗАКЛЮЧЕНИЕ

Были выполнены все шаги лабораторной работы. Мы научились создавать статические маршруты между маршрутизаторами в программе eNSP. А также настроили протокол OSPF на маршрутизаторах.